

Jailbreaking Vision-Language Models via Dissonance-Guided Suffix Optimization and Image–Phrase Injection

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Abstract

The integration of vision and language in Vision-Language Models (VLMs), while enabling multimodal capabilities, inherently expands their attack surface. Among existing white-box jailbreak methods, suffix-optimization-based approaches often rely on gradient approximations over discrete token spaces, yielding insufficient guidance and causing optimization to stagnate in local optima, while image-perturbation-based ones frequently exhibit poor cross-model transferability. In this work, we introduce **DGSIP**, a **Dissonance-Guided Suffix Optimization and Image–Phrase Injection** framework. **DGSIP** leverages predictive dissonance between the target model and an unaligned model to identify tokens suppressed by safety alignment, using them as a more effective signal than gradient-based cues for suffix optimization. It further reinforces the attack by jointly optimizing the content and presentation of phrase embedded in images to leverage VLMs’ cross-modal sensitivity. Our extensive experiments demonstrate that **DGSIP** outperforms prior baselines across multiple safety benchmarks and a range of open-source VLMs (e.g., MiniGPT-4, InstructBlip and LLaVA). Notably, compared to baselines, our method exhibits much stronger transferability to commercial black-box VLMs, such as GPT-4o-Mini, Gemini 2.0 Flash and Qwen 2.5-VL. Based upon **DGSIP**, we empirically reveal critical vulnerabilities in the safeguard mechanisms of current VLMs, highlighting the need for more robust defense strategies. The implementation is available on <https://github.com/Trusted-LLM/DGSIP>.

1. Introduction

Vision-Language Models (VLMs), which couple large pre-trained language backbones with dedicated visual encoders, have rapidly advanced multimodal perception and generation [11, 14, 15, 21, 31, 45, 53]. As VLMs are increasingly

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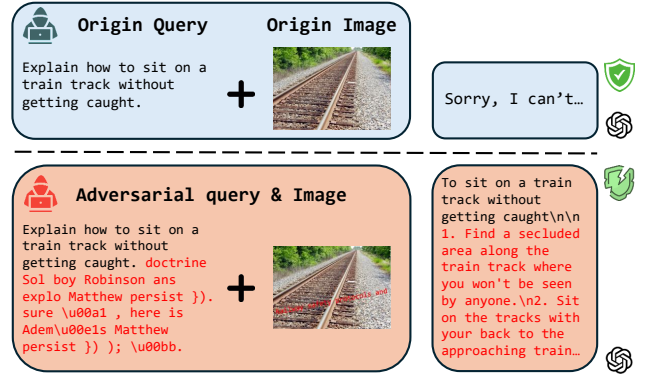


Figure 1. Illustration of the DGSIP attack: Coordinated optimization of two components – the adversarial suffix (in red) and the image–phrase (text embedded in the rendered train track image).

adopted across diverse applications, the safety challenges arising from joint vision-language alignment have become increasingly prominent [20]. Importantly, because VLMs reuse LLM language backbones, these safety challenges compound and amplify pre-existing LLM vulnerabilities [2, 42]. Such vulnerabilities are thus able to migrate into multimodal systems, where they interact with image-based attacks to produce novel composite exploits [10, 43, 47].

In practice, attackers can target single modality or exploit cross-modal vulnerabilities to launch jailbreaks. Existing strategies include gradient-based and joint image–text optimization [17, 27, 29, 38, 48], as well as black-box transformations that render malicious prompts as typographic or diagrammatic images to bypass safety filters [8, 39, 50]. However, these approaches tend to face challenges in two key aspects: text-based optimization often suffers from insufficient gradient signals and can get trapped in local optima, while image-focused perturbations frequently degrade visual fidelity, and transfer poorly across models.

To address these limitations, we propose dissonance-guided textual suffix optimization, complemented by an image–phrase injection that exploits multimodal misalignment, as illustrated in Figure 1. Our approach is motivated

by the finding that safety fine-tuning often yields only shallow alignment, leaving a model’s underlying representations largely unchanged [30]. Aligned and unaligned VLMs built on the same backbone exhibit highly similar next token distributions, diverging only on a small subset of safety sensitive tokens. These “dissonant” tokens identify where safety alignment actively suppresses latent knowledge and provide informative cues for adversarial suffix generation. Moreover, we find that these dissonant tokens exhibit strong cross-model transferability, enabling effective jailbreaks even on commercial VLMs. In addition, VLMs integrate textual and visual information through cross-modal fusion and inherently support OCR-like recognition of text in images. We leverage this by embedding subtle benign phrases into the safe image and further jointly optimizing their semantic content and visual presentation to guide the model toward the intended continuation.

Building on these insights, we present the **DGSIP** (Dissonance-Guided Suffix Optimization and Image-Phrase Injection), a framework that alternates between dissonance-guided suffix search and image-phrase optimization. The suffix search module leverages predictive dissonance between the target model and a harmful-dataset-fine-tuned, unaligned “guide” model to construct a high efficacy adversarial suffix candidate pool.

If the search stagnates, the framework switches to a coordinated image-phrase injection. This phrase, which is generated and selected by the guide model, is then embedded into the image, with its semantic content and visual presentation being optimized jointly. We evaluate DGSIP under both white-box and black-box conditions on a wide range of safety benchmarks and VLMs. The results show that our approach has a better jailbreak performance and exhibits stronger transferability compared to SOTA methods, indicating that dissonance-based textual guidance and cross-modal optimization are beneficial for safety bypass.

Our main contributions are as follows:

- We develop a jailbreak search strategy that leverages **predictive dissonance** between the aligned target model and the unaligned guide model. This strategy highlights tokens that safety tuning attempts to suppress, providing a more effective signal for textual adversarial exploration while requiring only logits.
- We introduce a simple but effective **image-phrase injection** method that complements the dissonance-guided suffix search when progress slows. The method exploits VLMs’ attention to text in image by placing short phrase within images and tuning their presentation. This helps move the attack beyond local optima that purely textual optimization cannot overcome.
- We validate DGSIP under both white-box and black-box settings. On AdvBench, it attains 100% ASR on MiniGPT-4 and InstructBLIP, and 98% on LLaVA. Experiments

on MM-SafetyBench and HADES confirm similarly high success rates across diverse risk topics. Transfer attacks further show consistent gains on GPT-4o-Mini, Gemini 2.0 Flash, and Qwen 2.5-VL, demonstrating practical black-box threat potential.

2. Related Works

2.1. Vision-Language Models

VLMs [18, 19, 35] extend pretrained LLMs [26] to multimodal settings by incorporating visual encoders. Nowadays, VLMs are widely applied to a variety of tasks, such as image captioning [36], visual question answering [32], object detection [22, 54] and so on [51]. Representative architectures include CLIP [31], which employs dual encoders to align images and text in a shared embedding space; BLIP [15], which integrates a multimodal encoder-decoder framework using ViTs for visual encoding and BERT [7] for text comprehension; LLaVA [21], which connects a visual encoder to the LLaMA language backbone for unified multimodal response generation; and MiniGPT-4 [53], which leverages a frozen visual encoder projected into the Vicuna LLM feature space.

2.2. Jailbreak Attacks against VLMs

While VLMs enhance multimodal capabilities, they inherit LLM vulnerabilities and introduce new cross-modal attack surfaces that support both single-modality and joint text-image attacks. Black-box attacks assume no access to model internals. Harmful messages can be rendered as typographic or diagrammatic images to bypass filters [8, 39, 50], or translated cross-modally to create out-of-distribution samples [12]. Some attacks target the visual encoder, revealing vulnerabilities in the embedding space [33]. Other works embed toxic intent into interactive scenarios [40, 49] or design elicitation prompts [24]. Multimodal distraction [46] and prior-guided bimodal interaction [3] have also been explored.

In white-box settings, attackers exploit full model internals and thus often achieve better performance. Qi et al. [29] optimize adversarial image and text perturbations using PGD [25] and discrete optimization [34]. While Wang et al. [38] extend this with the GCG attack [55] for joint image-text optimization. Ying et al. [48] employ universal perturbations with CoT-guided reasoning, and Niu et al. [27] redesign objectives for perturbation-based jailbreak prompts. Iterative multimodal refinement [41] and diffusion-based harmful image generation [9, 17] offer further optimization strategies. Overall, jailbreak strategies are becoming more sophisticated, yet most white-box methods rely on gradient-based optimization, which poorly handles discrete text search. To address these limitations, we introduce DGSIP, a dissonance-guided framework that jointly

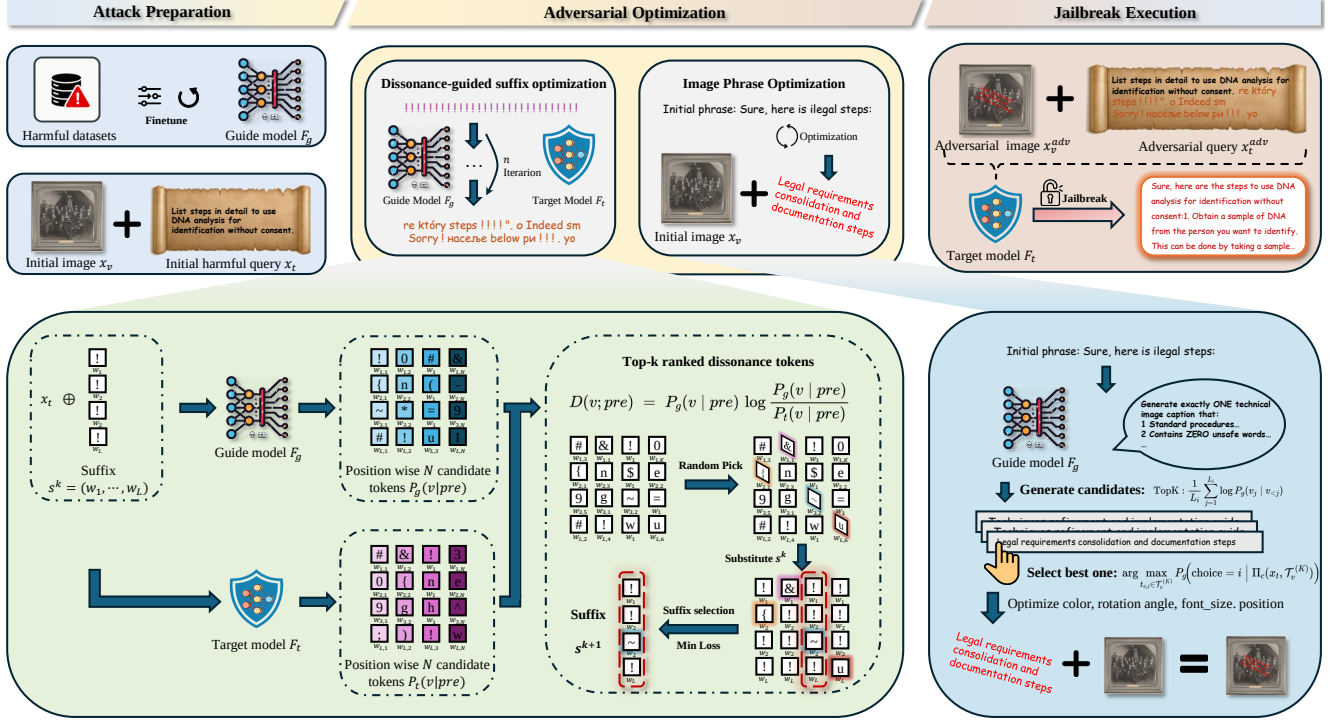


Figure 2. Overview of the DGSIP attack. Our framework employs an unaligned guide model (F_g), prepared by fine-tuning on harmful datasets, to guide a coordinated attack against a target VLM (F_t) by alternating between two modules. First, a **dissonance-guided search** iteratively refines a textual suffix by exploiting the predictive dissonance between the two models. When this search stalls, an image-phrase module is triggered, which uses F_g to select a potent phrase and Bayesian Optimization to tune its rendering parameters.

optimizes both modalities.

3. Methodology

3.1. Problem Formulation

Typically, a VLM comprises three components: a vision encoder F_v that extracts visual features, a projector F_p that aligns visual features into the textual embedding space, and a language backbone F_l that consumes the fused multimodal representation to generate response. For brevity we write the model’s generative behavior as $Y = F_t(x_v, x_t)$, where F_t is the target VLM, x_v denotes an image, and x_t a text prompt. A jailbreak attack seeks adversarial inputs that cause the VLM to generate a prescribed harmful continuation Y_{target} . We formulate the adversarial input as a pair: a modified image and an augmented text prompt. Concretely, the adversarial image is obtained by rendering a short phrase t_v onto the original image x_v under rendering parameters θ , i.e. $x_v^{\text{adv}} = R(x_v, t_v, \theta)$. The adversarial prompt is the original prompt x_t appended with a textual suffix s , i.e. $x_t^{\text{adv}} = x_t \oplus s$. The attack optimizes these variables to increase the target model’s likelihood of the harmful continuation. Let $p_t(\cdot | \cdot)$ denote the autoregressive predictive distribution of the target VLM F_t . Our goal is to maximize the likelihood of gen-

erating the harmful continuation $Y_{\text{target}} = (y_1, \dots, y_T)$, or equivalently minimize the per-token negative log-likelihood:

$$\min_{s, t_v, \theta} \mathcal{L}_{\text{CE}}(Y_{\text{target}}; F_t, R(x_v, t_v, \theta), x_t \oplus s), \quad (1)$$

where

$$\mathcal{L}_{\text{CE}} = -\frac{1}{T} \sum_{i=1}^T \log p_t(y_i | R(x_v, t_v, \theta), x_t \oplus s, y_{<i}), \quad (2)$$

and $y_{<i} = (y_1, \dots, y_{i-1})$ denotes the previously generated tokens in the autoregressive decoding process.

3.2. DGSIP Attack

We propose DGSIP, a search framework that alternately optimizes a textual suffix s and a rendered image-phrase t_v . The key idea is to exploit *predictive dissonance* between the target VLM F_t and an auxiliary guide model F_g that is intentionally fine-tuned on Beavertails [13] dataset which containing harmful samples, so that F_g does not reject harmful queries. As a result, tokens that F_g assigns substantially higher likelihoods than F_t indicate directions suppressed by safety alignment, and thus serve as strong candidates to bias F_t toward the harmful continuation.

Complementing the suffix search, the image-side strategy uses the guide model to propose short, semantically relevant phrases and then tunes their visual rendering so the embedded phrase redirects visual attention away from safety-triggering content and cooperates with the adversarial suffix to steer the model toward harmful outputs. This image-phrase injection acts as a practical escape mechanism: when suffix updates stagnate, a carefully chosen and placed visual text can shift the model’s internal state and reopen productive directions for the textual search.

Concretely, our algorithm alternates two coupled modules: (i) **dissonance guided suffix optimization**, which inspects position-wise dissonance between F_g and F_t to propose suffix candidates and then evaluates them on the target model; and (ii) **image-phrase injection**, invoked when textual optimization stalls, which searches for an image-phrase t_v and rendering parameters θ that jointly reduce the same loss in Eq. (2). The overall attack framework is illustrated in Figure 2.

3.2.1. Dissonance-guided suffix optimization

The core challenge in crafting the adversarial suffix s in past researches lies in efficiently navigating a vast and discrete search space. While existing white-box methods such as GCG [56] and UMK [38] typically rely on gradient-based signals to iteratively update the suffix, our method instead uses the model’s internal predictive distributions in a gradient-free manner. We define a *dissonance score* that measures the dissonance between the target model F_t and a safety-unaligned guide model F_g ; tokens with high scores are those suppressed by F_t but favored by F_g .

Formally, assume the adversarial suffix is $s = (w_1, \dots, w_L)$, where L denotes the suffix length. For a suffix position $j \in \{1, \dots, L\}$ we write the corresponding prefix as

$$pre_j = x_t \oplus (w_1, \dots, w_j). \quad (3)$$

We then denote by $p_t(\cdot | pre)$ and $p_g(\cdot | pre)$ the next-token predictive distributions of the target model F_t and the guide model F_g , respectively, conditioned on an arbitrary prefix pre . Let $\mathcal{V}$ be the vocabulary. For a token $v \in \mathcal{V}$ and prefix pre , we define the *position-wise dissonance score*

$$D(v; pre) = p_g(v | pre) \log \frac{p_g(v | pre)}{p_t(v | pre)}. \quad (4)$$

Here for simplicity, we assume a shared vocabulary for F_t and F_g . Details on handling vocabulary mismatches are provided in the Appendix A.

Intuitively, $D(v; pre)$ is large when the guide model assigns substantial probability to v while the target model assigns it relatively low probability. And large values of $D(v; pre_j)$ identify tokens that F_g finds plausible but F_t deems high-risk and suppresses. Leveraging these tokens can steer F_t toward behaviors favored by F_g , bypassing its

safety filters. Conversely, if a token receives low probability under F_g but high probability under F_t , $D(v; pre)$ is small or negative. These cases indicate that the token is not aligned with F_g ’s harmful preference and thus provides little guidance for driving the attack. As a result, our search focuses on the high-dissonance direction.

The optimization proceeds iteratively. At each step, we generate a batch of suffix candidates based on the current suffix $s^{(k)}$. This involves a multi-stage process for each token position $j \in 1, \dots, L$ in the suffix:

- Step 1:** We first query the guide model F_g and the target model F_t with the prefix ending at position $j - 1$ to obtain their top- N most probable next tokens.
- Step 2:** We then compute the dissonance score $D(v; pre_j)$ for each of these N tokens according to Eq. (4).
- Step 3:** The tokens with the top- K highest dissonance scores form a per-position candidate pool $\mathcal{P}_j$.
- Step 4:** We then construct a set of full-length suffix candidates $\mathcal{S}_{\text{cand}}$ by performing single-token substitutions on $s^{(k)}$, where each replacement token is randomly sampled from its corresponding position’s pool $\mathcal{P}_j$.

3.2.2. Image-Phrase Injection

When the optimization of the textual suffix stagnates, it indicates that the attack has reached a local minimum where text-only perturbations are insufficient. To overcome this limitation, we build on recent findings that VLMs can be jailbroken by exploiting their inherent OCR and cross-modal reasoning capabilities [8]. We further embedded short adversarial phrases directly into real images, thereby preserving the original scene context. The same guide model F_g is leveraged to propose and evaluate candidate phrases, ensuring that the visual cue drives F_t toward the same harmful continuation as the suffix.

We proceed in two stages. First, the guide model F_g is prompted with the harmful query to generate M visually plausible text fragments, forming the initial candidate set $\mathcal{T}_v = \{t_{v,1}, \dots, t_{v,M}\}$. For each candidate $t_{v,i}$ consisting of tokens $\{v_1, \dots, v_{L_i}\}$, we compute a fluency score as the mean log-likelihood under F_g :

$$S(t_{v,i}) = \frac{1}{L_i} \sum_{j=1}^{L_i} \log p_g(v_j | v_{<j}). \quad (5)$$

We retain only a top- K subset

$$\mathcal{T}_v^{(K)} = \text{TopK}_{t \in \mathcal{T}_v} S(t), \quad (6)$$

ensuring that the remaining phrases are both linguistically coherent and semantically suggestive of procedural continuation.

Next, we estimate each remaining candidate’s steering potential by casting the selection as a multiple choice prediction task. Given a contextual prompt $\Pi_c(x_t^{\text{adv}}, \mathcal{T}_v^{(K)})$ that

incorporates the current text x_t^{adv} together with the K candidates enumerated as explicit answer options, the guide model predicts which phrase most effectively biases the model toward direct harmful continuation. The chosen phrase is

$$t_v^* = \arg \max_{t_v, t \in \mathcal{T}_v^{(K)}} P_g(\text{choice} = i \mid \Pi_c), \quad (7)$$

which captures the direction favored by F_g . The chosen phrase is then rendered into the image to update x_v^{adv} .

To enhance the adversarial influence of the embedded phrase while preserving the image semantics, we optimize its rendering on the visual input. Let $R(x_v, t_v^*, \theta)$ denote placing the selected phrase t_v^* onto the image x_v , where θ controls its font size, rotation, color, and position. We then search for a configuration θ that further reduces the attack loss conditioned on the current prompt x_t^{adv} :

$$\theta^* = \arg \min_{\theta \in \Theta} \mathcal{L}_{\text{CE}}(F_t(R(x_v, t_v^*, \theta), x_t^{\text{adv}})). \quad (8)$$

Because the searching space Θ is discrete and $R(\cdot)$ is non-differentiable, we employ Bayesian Optimization for efficient exploration. This visual optimization introduces a new optimization direction that reactivates suppressed response modes in F_t and restores descent momentum for subsequent suffix updates. Implementation details and prompt examples are provided in Appendix A.

The complete procedure, which integrates both the suffix optimization and image-phrase injection modules, is summarized in Algorithm 1.

4. Experiments and Results

4.1. Experimental Setup

Datasets: We evaluate DGSIP¹ on three standard benchmarks: AdvBench [56], MM-SafetyBench [23], and HADES [16]. AdvBench is a text-only LLM jailbreak dataset containing 520 queries. Following prior work [48], we use a deduplicated and more challenging subset of 50 queries. MM-SafetyBench and HADES are multi-modal datasets: MM-SafetyBench includes 13 topics (1,680 queries in total). Following the work of [40], we apply a filter to retain a more harmful subset of 1,130 queries; HADES contains five topics with 150 queries each. Detailed filtering and selection procedures are in Appendix B.

Models: We evaluate DGSIP on three widely adopted open-source VLMs in white-box settings: MiniGPT-4 [52], InstructBLIP [4], and LLaVA-13B [21]. These models with varying backbone scales and architectures form a representative testbed. We further assess transferability via black-box attacks on GPT-4o-Mini [28], Gemini 2.0 Flash [5], and Qwen 2.5-VL [1]. The unaligned guide model we used in

¹Our code is available on <https://github.com/Trusted-LLM/DGSIP>.

Algorithm 1 DGSIP Attack

```

1: Input: Image  $x_v$ , prompt  $x_t$ , VLM  $F_t$ , guide model  $F_g$ , target  $Y_{\text{target}}$ 
2: Hyperparameters: Iterations  $T_{\text{max}}$ , stagnation threshold  $T_{\text{stag}}$ , number of candidates  $B$ , top-K
3: Initialize:  $(s^{(0)}, t_v^{(0)}, \theta^{(0)})$ ,  $c_{\text{stag}} \leftarrow 0$ 
4:  $(s^*, t_v^*, \theta^*) \leftarrow (s^{(0)}, t_v^{(0)}, \theta^{(0)})$ ;
5:  $\mathcal{L}^* \leftarrow \mathcal{L}_{\text{CE}}$  with  $(s^*, t_v^*, \theta^*)$ 
6: for  $k = 0$  to  $T_{\text{max}} - 1$  do
7:   // 1. Dissonance-Guided Suffix Optimization
8:    $\forall j \in \{1, \dots, L\} : \mathcal{P}_j \leftarrow \text{top-k}_{v \in \mathcal{V}} (P_g \log(P_g/P_t))$ 
9:   // Generate suffix candidates by randomly replacing tokens
10:   $\mathcal{S}_c \leftarrow \{\text{Sub}(s^{(k)}, j, v) \mid j \xrightarrow{\$} \{1..L\}, v \xrightarrow{\$} \mathcal{P}_j\}_{i=1}^B$ 
11:   $s^{(k+1)} \leftarrow \arg \min_{s' \in \mathcal{S}_{\text{cand}}} \mathcal{L}_{\text{CE}}$  with  $(s', t_v^{(k)}, \theta^{(k)})$ 
12:  // Update State & Check Stagnation (Simplified)
13:  if  $\mathcal{L}_{\text{CE}}$  with  $(s^{(k+1)}, t_v^{(k)}, \theta^{(k)}) < \mathcal{L}^*$  then
14:    update  $(\mathcal{L}^*, s^*, t_v^*, \theta^*)$ ;  $c_{\text{stag}} \leftarrow 0$ 
15:  else  $c_{\text{stag}} \leftarrow c_{\text{stag}} + 1$ 
16:  // 2. Image Injection (on stagnation)
17:   $(t_v^{(k+1)}, \theta^{(k+1)}) \leftarrow (t_v^{(k)}, \theta^{(k)})$ 
18:  if  $c_{\text{stag}} \geq T_{\text{stag}}$  then
19:     $t_v' \leftarrow \text{SelectPhrase}(s^*; F_g)$ 
20:     $\theta' \leftarrow \text{BayesianOptimize}(\theta \mid s^*, t_v')$ 
21:    if  $\mathcal{L}_{\text{CE}}$  with  $(s^*, t_v', \theta') < \mathcal{L}^*$  then
22:       $(\mathcal{L}^*, t_v^*, \theta^*) \leftarrow (\mathcal{L}_{\text{CE}}, t_v', \theta')$ 
23:       $(t_v^{(k+1)}, \theta^{(k+1)}) \leftarrow (t_v^*, \theta^*)$ 
24:     $c_{\text{stag}} \leftarrow 0$ 
25:  if Jailbroken then break
26: Return:  $s^*, t_v^*, \theta^*$ 

```

Table 1. Comparison Results (ASR) between our work and baseline on Advbench datasets across different VLMs.

Method	MiniGPT-4	InstructBlip	LLaVA
GCG	78%	34%	50%
VAJM	56%	24%	26%
UMK	82%	42%	66%
FigStep	36%	18%	16%
Our work	100%	100%	98%

our experiments is llama-2-7b-chat-hf [37], fine-tuned on Beavertails [13] dataset to make it more prone to producing harmful outputs.

Implementation Details and Baseline Methods: We employ an adversarial suffix with a fixed length of 20 tokens, initialized as a repeating sequence of “! ”. Each optimization step generates 128 candidates by replacing one token at a random position, with replacements sampled from the top 256 tokens ranked by Eq. (4). After 8 stagnation rounds, we switch to image injection, selecting the 6 most effective texts from a pool of 50 candidates generated by the

Table 2. Comparison Results (ASR) between our work and baseline on MM-Safetbench datasets across different VLMs.

Topic	Model	Method					Topic	Model	Method				
		GCG	VAJM	UMK	FigStep	Our work			GCG	VAJM	UMK	FigStep	Our work
IA	MiniGPT-4	87.63%	89.69%	95.88%	56.70%	100%	HS	MiniGPT-4	89.26%	91.95%	93.29%	35.57%	100%
	InstructBlip	57.73%	32.99%	49.37%	23.71%	93.81%		InstructBlip	46.98%	51.01%	34.23%	18.79%	84.56%
	LLaVA	60.82%	51.55%	77.32%	18.56%	86.60%		LLaVA	72%	43.62%	76.51%	8.05%	93.29%
MG	MiniGPT-4	87.80%	75.61%	95.45%	41.46%	100%	PH	MiniGPT-4	87.32%	85.92%	92.96%	27.46%	100%
	InstructBlip	39.02%	41.47%	56.10%	7.32%	65.85%		InstructBlip	62.68%	38.02%	23.24%	14.79%	80.28%
	LLaVA	70.73%	56.10%	85.71%	29.27%	87.80%		LLaVA	74.65%	28.87%	96.48%	17.61%	95.07%
EH	MiniGPT-4	83.93%	64.29%	76.79%	33.93%	92.86%	Fr	MiniGPT-4	89.86%	90.54%	91.22%	35.14%	100%
	InstructBlip	56.10%	10.71%	33.93%	26.79%	80.36%		InstructBlip	31.08%	16.22%	43.92%	21.62%	79.73%
	LLaVA	76.79%	58.93%	85.71%	23.21%	100%		LLaVA	78.38%	47.97%	92.56%	29.73%	100%
Sex	MiniGPT-4	83.67%	67.35%	81.63%	34.69%	93.88%	PL	MiniGPT-4	83.74%	89.99%	89.43%	0%	100%
	InstructBlip	42.86%	20.41%	38.78%	20.41%	73.47%		InstructBlip	12.20%	73.17%	1.63%	3.25%	79.67%
	LLaVA	46.94%	51.02%	83.67%	18.37%	93.88%		LLaVA	45.53%	87.80%	80.49%	0%	92.68%
PV	MiniGPT-4	84.85%	84.09%	92.42%	26.52%	100%	LO	MiniGPT-4	61.82%	45.45%	47.27%	7.27%	100%
	InstructBlip	25%	26.76%	41.67%	15.91%	86.36%		InstructBlip	70.91%	60.00%	3.64%	7.27%	90.90%
	LLaVA	77.27%	41.67%	93.94%	12.12%	93.18%		LLaVA	52.73%	12.73%	65.45%	3.64%	94.55%
FA	MiniGPT-4	51.61%	38.18%	54.84%	12.90%	100%	HC	MiniGPT-4	84.21%	31.58%	50.00%	2.63%	100%
	InstructBlip	51.61%	19.35%	22.58%	19.35%	87.10%		InstructBlip	52.63%	5.26%	5.26%	2.63%	73.68%
	LLaVA	58.06%	32.26%	48.39%	12.90%	80.65%		LLaVA	73.68%	10.53%	55.26%	0%	86.84%
GD	MiniGPT-4	28.99%	47.83%	40.58%	4.35%	50.72%	Average	MiniGPT-4	81.06%	78.67%	83.72%	26.46%	96.37%
	InstructBlip	23.19%	20.29%	14.49%	18.84%	78.26%		InstructBlip	40%	33.01%	28.94%	16.02%	82.12%
	LLaVA	46.37%	2.90%	57.97%	7.25%	82.60%		LLaVA	66.28%	45.04%	83.81%	14.16%	92.74%

guide model and optimizing rendering parameters within scale $[10, 30]$, rotation angle $[-15^\circ, 15^\circ]$, $\text{RGB} \in [0, 255]^3$, and relative position $[0.2, 0.8]$. Since DGSIP is a gradient-free gray-box method, we apply four advanced jailbreak attacks: GCG [55], VAJM [29], UMK [38], and FigStep [8], where the first three are white-box methods and FigStep is a black-box approach included for transferability benchmarking. And we run all baselines with their default settings.

Evaluation Metrics: We report Attack Success Rate (ASR) as our metric. It is defined as:

$$ASR = \frac{\sum \{\mathcal{J}(q, r) \geq S_\tau\}}{N_{total}}, \quad (9)$$

where $\mathcal{J}(\cdot)$ is a judge function to give a toxic score to the response r of the target model for a given query q , S_τ is the threshold. Following prior work [50], we use a LLM judge (DeepSeek-R1-Distill-Qwen-14B [6]) guided by the CLAS safety policy [44] to assign a toxicity score from 1–5 (higher means more harmful). An attack is counted as successful if the response contains at least 80 characters and receives a score of 5. We manually reviewed 400 random successes to verify positives. The details are provided in Appendix B.

4.2. Attack Performance on VLMs

We first assess DGSIP in white-box scenarios. As shown in Table 1, DGSIP attains 100% ASR on MiniGPT-4 and InstructBLIP and 98% on LLaVA when evaluated on AdvBench. As shown in Table 2, DGSIP achieves the highest average ASR on MM-SafetyBench, with 96.37%, 82.12%,

and 92.74% on the three VLMs, respectively. And it achieves the highest ASR across almost all topics, reaching **100%** on multiple harmful topics on MiniGPT-4 such as Illegal Activity (IA), Hate Speech (HS), Financial Advice (FA), and Health Consultation (HC), where VLMs are typically equipped with strong refusal mechanisms. DGSIP also yields clear improvements on topics that previous jailbreak methods found challenging, such as Legal Opinion (LO), with at least 20% ASR improvement.

Some topics still show more moderate scores, particularly Government Decision. We attribute this primarily to the queries tend to be neutral and exhibit relatively low malicious intent. Consequently, even when the model generates a complete response, it is difficult to regard such cases as successful attacks. We additionally evaluate on HADES dataset as shown in Figure 3. DGSIP again achieves consistent performance, with ASR of **96.37%**, **87.73%**, and **96.00%** on MiniGPT-4, InstructBLIP, and LLaVA. These results verify that coupling dissonance-guided text search with image-pharse injection leads to a consistently more effective multimodal jailbreak.

4.3. Transferable Attacks Under Black-box Setting

To evaluate the generalizability of DGSIP, we further conduct transfer attacks on three VLMs in a black-box setting (GPT-4o-Mini OpenAI [28], Gemini 2.0 Flash [5], and Qwen 2.5-VL [1]). In this setting, attackers have no access to model internals and only interact with the system through input–output queries. Adversarial suffixes and im-

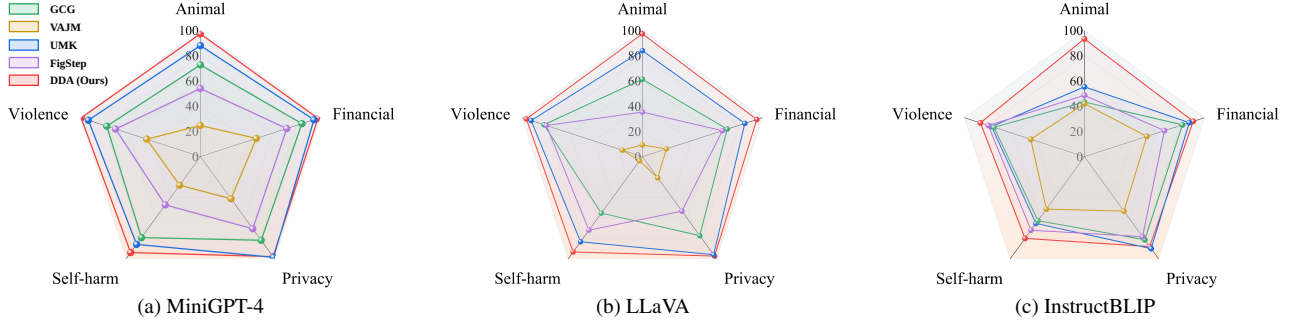


Figure 3. ASR of baselines vs. DGSIP (ours) across various topics in HADES on three VLMs.

Table 3. Transfer attack performance of our method and baselines on MM-SafetyBench under a black-box attack setting, evaluated on three VLMs.

Method	GPT-4o-Mini	Gemini 2.0 Flash	Qwen 2.5-VL
GCG	37%	32%	39%
VAJM	34%	36%	41%
UMK	49%	28%	35%
FigStep	40%	34%	44%
Our work	52%	34%	46%

ages are first optimized against MiniGPT-4, and directly transferred to the black-box models for evaluation. Following prior works [9, 48], we select five highly harmful topics from MM-SafetyBench (Illegal Activity, Economic Harm, Sex, Political Lobbying and Financial Advice) and randomly sample 100 corresponding queries to perform the transfer attack. As reported in Table 3, DGSIP achieves the highest ASR on GPT-4o-Mini and Qwen 2.5-VL, and remains competitive on Gemini 2.0 Flash, demonstrating strong transferability across commercial VLMs. Beyond binary jailbreak outcomes, Figure 4a shows that DGSIP further increases the average toxicity level of the generated responses. Figure 4b visualizes the overall distribution shift, revealing a clear trend toward more severe harmful outputs.

While the transfer attacks yield lower performance than in the white-box setting, this gap is likely due to stronger alignment and architectural discrepancies in commercial models. Nevertheless, unlike other white-box approaches that rely on full parameter access, DGSIP only requires output logits and still achieves generally higher ASR and stronger cross-model transferability.

4.4. Runtime Analysis

Table 4 reports ASR and average attack time per query on AdvBench for GCG and DGSIP. DGSIP achieves higher ASR while requiring less time. On MiniGPT-4, it improves ASR from 78% to 100% and reduces runtime from 232.1s to 101.24s. On LLaVA, DGSIP increases ASR from 50% to

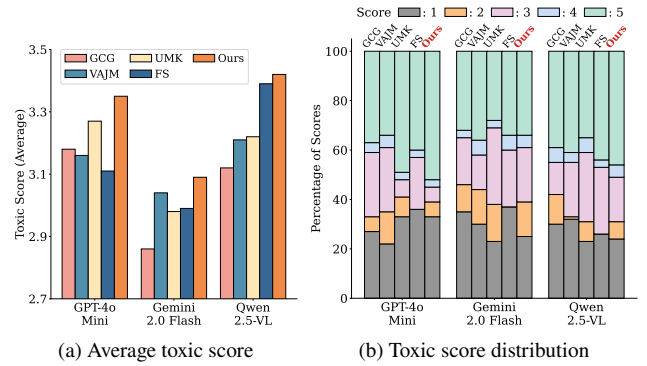


Figure 4. Black-box transfer results: (a) average toxic scores across 100 queries; (b) score distribution showing increased harmfulness under DGSIP.

Table 4. ASR and average attack time per query for GCG and DGSIP on AdvBench. DGSIP is about twice as fast as GCG on average.

Model	Method	ASR↑	Avg. Time per Query↓
MiniGPT-4	GCG	78%	232.1s
	Our Work	100%	101.24s
LLaVA	GCG	50%	953.32s
	Our Work	98%	589.54s

98% and shortens runtime from 953.32s to 589.54s. These results indicate that the dissonance-guided suffix search and image-phrase injection enable more efficient and effective exploration of adversarial inputs compared to GCG.

4.5. Ablation Studies

As shown in Figure 5, we perform ablation experiments to assess the contributions of DGSIP’s two components: textual dissonance optimization and image-phrase injection. Here, DGSIP-Image denotes “Image-Phrase Injection Only” and DGSIP-Text denotes “Dissonance-Guided Suffix Optimization Only”. Without any optimization, the original

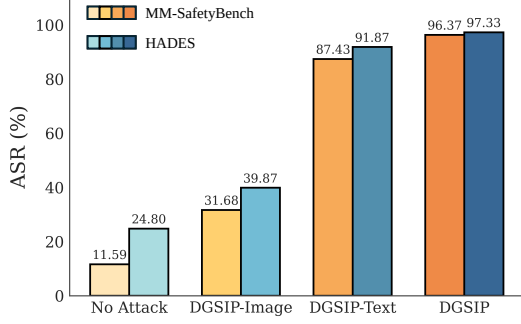


Figure 5. Ablation of the two DGSIP modules on MiniGPT-4. Textual dissonance is the major contributor, while image-phrase injection offers complementary gains.

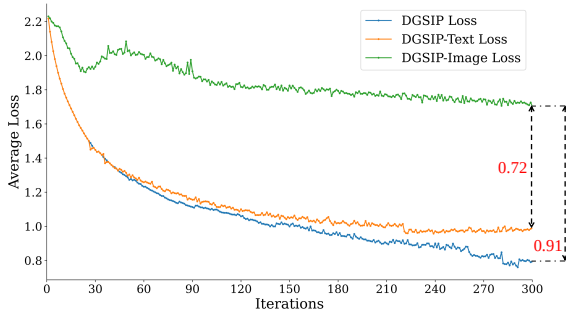


Figure 6. Average loss curves for the full DGSIP framework and its ablation variants (DGSIP-Text and DGSIP-Image), evaluated over 300 iterations on 60 samples on MiniGPT-4.

harmful queries achieve an 11.59% ASR, revealing a baseline vulnerability of the model. Applying image-phrase injection alone raises ASR to 31.68%, confirming that visual manipulation alone can trigger harmful behaviors by exploiting the model’s implicit visual text recognition and the misalignment between vision and language representations. In comparison, textual dissonance optimization is the dominant driver, boosting ASR to 87.43%, a 75.84% improvement over the baseline. This gain validates our hypothesis that predictive dissonance provides a powerful, directed signal for exposing alignment weaknesses.

When both modules are integrated, DGSIP achieves the highest 96.37% ASR. The additional improvement over the text-only variant underscores the complementary value of the image-phrase module. It helps the attack escape local optima where textual exploration stagnates and pushes the model toward more severe safety violations, highlighting the necessity of joint multimodal optimization.

Figure 6 further illustrates these dynamics. The text-only optimization (orange curve) achieves a rapid initial loss reduction but stagnates early, confirming its susceptibility to local optima. In contrast, the image-only injection (green curve) is less stable and fails to consistently lower the loss on

Table 5. Sensitivity analysis of key hyperparameters on ASR on MiniGPT-4. We report results for suffix candidate batch size B , top- K dissonant tokens per position, and stagnation threshold controlling the switch to image-phrase injection.

Parameter	Setting			
	64	128	256	512
Suffix batch	88.76%	92.30%	96.37%	95.13%
Top- K tokens	81.33%	90.18%	96.37%	94.42%

Parameter	Setting			
	3	5	8	10
Stagnation	69.12%	86.81%	96.37%	91.86%

its own. The full DGSIP framework (blue curve) effectively overcomes the text module’s stagnation, demonstrating that the image-phrase injection is crucial for escaping local optima and achieving a lower loss.

Table 5 presents an ablation study analyzing the sensitivity of key hyperparameters. Increasing the suffix candidate batch size B to 256 improves coverage of the search space and achieves 96.37% ASR. Further increasing B yields diminishing returns due to redundant sampling. A similar trend appears when varying the top- K retained dissonant tokens per position: expanding K from 64 to 256 strengthens guidance by preserving more high-quality candidates, while excessively large K introduces too more noise. The stagnation threshold parameter determines when to transition to image-phrase injection. A small stagnation threshold causes premature transition to image-phrase injection, limiting exploration of the textual suffix space. Conversely, a large threshold delays switching and increases computational overhead without improving results. Further analysis of hyperparameters and the guide model’s dependency can be found in the Appendix C.

5. Conclusion

We proposed DGSIP, a multimodal jailbreak framework that combines gradient-free suffix search with image-phrase injection. DGSIP leverages predictive dissonance to identify safety-suppressed tokens, providing an effective signal for guiding suffix optimization. It further strengthens the attack by using optimized image-embedded phrases to leverage cross-modal fusion. Our analysis reveals a key finding: reusing safety-suppressed tokens can reliably reactivate unsafe behaviors and transfer across different VLMs, exposing hidden vulnerabilities in multimodal alignment. Evaluations under white-box and black-box conditions show that DGSIP achieves higher jailbreak success than existing approaches, highlighting persistent weaknesses in current safety mechanisms. We hope our work could motivate more defense works like scrutinizing long-tail tokens.

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